

2

NUMBER AND ALGEBRA, MEASUREMENT AND SPACE

SURDS AND PYTHAGORAS' THEOREM

Pythagoras was an ancient Greek mathematician who lived in the 5th century BCE. The theorem (or rule) which carries his name was well-known before his time, but Pythagoras may have been the first to prove it. Over 300 proofs for the theorem are known today. Pythagoras' theorem is the most famous mathematical formula, and it is still used today in architecture, engineering, surveying and astronomy.



Chapter outline

	Working mathematically				
2.01 Finding the hypotenuse	U	F	PS	R	C
2.02 Surds	U	F		R	C
2.03 Simplifying surds*	U	F		R	C
2.04 Adding and subtracting surds*	U	F		R	
2.05 Multiplying and dividing surds*	U	F		R	
2.06 Finding a shorter side	U	F	PS	R	
2.07 Pythagoras' theorem problems	U	F	PS		C
2.08 Testing for right-angled triangles	U	F		R	C
2.09 Pythagorean triads	U	F		R	C

***PATH**

Wordbank

converse A rule or statement turned back-to-front; the reverse statement

hypotenuse The longest side of a right-angled triangle; the side opposite the right angle

irrational number A number such as π or $\sqrt{2}$ that cannot be expressed as a fraction

Pythagoras An ancient Greek mathematician who discovered an important formula about the sides of a right-angled triangle

Pythagorean triad A set of three numbers that follow Pythagoras' theorem, such as {3, 4, 5}.

surd A square root (or other root) whose exact value cannot be found

theorem Another name for a formal rule or formula

Quiz
Wordbank 2

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

2.01, 2.06 Pythagoras' theorem 1 • Pythagoras' theorem 2
2.02 Surds and irrational numbers
2.03 Simplifying surds 1 • Simplifying surds 2
2.04 Adding and subtracting surds
2.07 Pythagoras' theorem problems
2.08 Testing for right-angled triangles

***Twig* videos (3):**

2.01 Irrational numbers: Pythagoras • Proving Pythagoras • Building the pyramids

Quizzes (6):

- Wordbank 2
- SkillCheck 2
- Mental skills 2A
- Mental skills 2B
- Language of maths 2
- Test yourself 2

Skillsheets (2):

2.01 Pythagoras' theorem • Spreadsheets

Worksheets (10):

2.01 Pythagoras' discovery • A page of right-angled triangles • Pythagoras 1
2.02 Surds on the number line
2.03 Surds
2.06 Finding an unknown side
2.07 Applications of Pythagoras' theorem
2.08 Testing for right-angled triangles
2.09 Pythagorean triads
Mind map Surds and Pythagoras' theorem

Puzzles (6):

2.03 Simplifying surds
2.04 Surds code puzzle
2.07 Pythagoras 1 • Pythagoras 2 • Pythagorean two-step problems
Language of maths Pythagoras' theorem find-a-word

Technology (3):

2.01 Pythagoras' theorem
2.03 Simplifying surds quiz
2.06 Pythagoras' theorem
2.09 Pythagorean triples

Spreadsheets (3):

2.01, 2.06 Pythagoras' theorem
2.03 Simplifying surds quiz
2.09 Pythagorean triples

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- investigate irrational numbers and surds
- (PATH) describe real numbers and the properties of surds
- (PATH) simplify, add, subtract, multiply and divide surds
- solve problems involving Pythagoras' theorem, writing the answers in decimal or surd form
- test whether a triangle is right-angled
- investigate Pythagorean triads

Quiz
SkillCheck 2

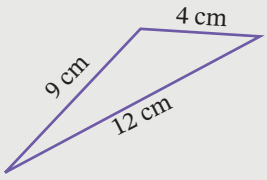
SkillCheck ANSWERS ON P. 613

1 Evaluate each expression.

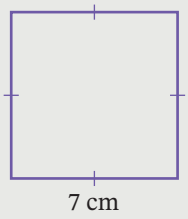
a 4^2 **b** 10.3^2 **c** $3^2 + 5^2$
d $8^2 - 6^2$ **e** $\sqrt{49}$ **f** $\sqrt{121}$

2 Find the perimeter of each shape.

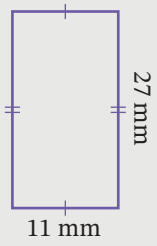
a



b



c



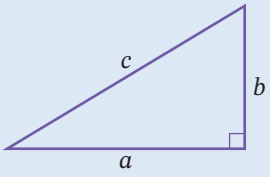
3 Select the square numbers from the following list of numbers.

44 81 25 100 75 72 16 50 64 32

Finding the hypotenuse 2.01

Pythagoras' theorem

For any right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other 2 sides.



$$c^2 = a^2 + b^2$$

that is,

$$(\text{hypotenuse})^2 = (\text{shorter side 1})^2 + (\text{shorter side 2})^2$$

Example 1

Find the value of c in this triangle.



Solution

We want to find the length of the hypotenuse.

Using Pythagoras' theorem:

$c^2 = 9^2 + 40^2$

$= 1681$

$c = \sqrt{1681}$ Use the square root to find c

$= 41$

An answer of $c = 41$ looks reasonable because:

- the hypotenuse is the longest side
- from the diagram, the hypotenuse looks a little longer than the side that is 40 cm

Worksheets
Pythagoras' discovery
A page of right-angled triangles

Skillsheet
Pythagoras' theorem

Video
Pythagoras' theorem 1

Technology
Pythagoras' theorem

Spreadsheet
Pythagoras' theorem

Videos
Irrational numbers: Pythagoras
Proving Pythagoras
Building the pyramids

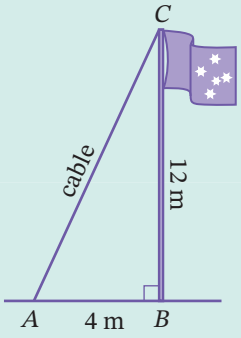
Video
Pythagoras' theorem 2

Example 2

Find the length of the cable supporting this flagpole:

a as a surd

b correct to one decimal place.



Solution

a $AC^2 = 12^2 + 4^2$

$= 160$

$AC = \sqrt{160}$ m — This is the answer as a surd (in $\sqrt{\ }$ form).

b $AC = \sqrt{160}$ — From part **a**

$= 12.6491...$

≈ 12.6 m — Rounded to one decimal place.

Example 3

In $\triangle PQR$, $\angle P = 90°$, $PQ = 25$ cm and $PR = 32$ cm. Sketch the triangle and find the length of the hypotenuse, correct to one decimal place.

Solution

$p^2 = 32^2 + 25^2$ — p is the side opposite $\angle P$ and is also the hypotenuse

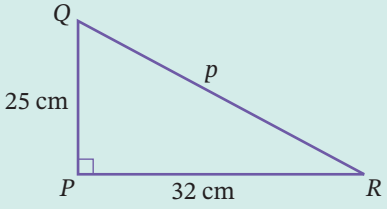
$= 1649$

$p = \sqrt{1649}$

$= 40.607\,881\,01...$

≈ 40.6

$\therefore$ The length of the hypotenuse is 40.6 cm.

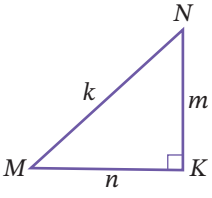


EXERCISE 2.01 ANSWERS ON P. 613

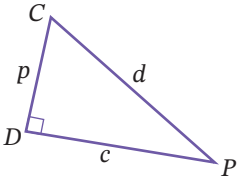
Finding the hypotenuse U F PS R C

1 Write Pythagoras' theorem for each triangle. C

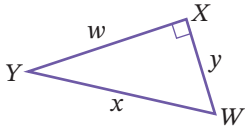
a



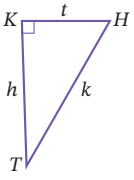
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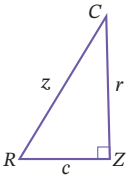
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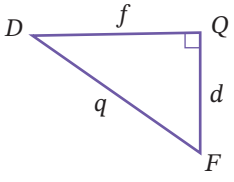
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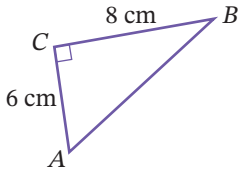


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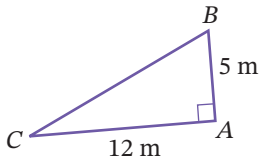
2 Find the length of the hypotenuse in each triangle.

EXAMPLE 1

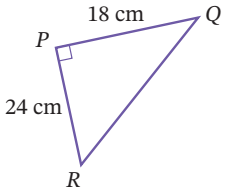
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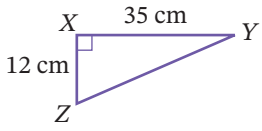
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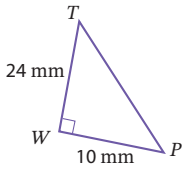
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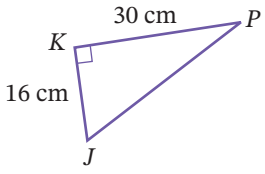
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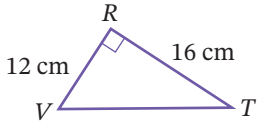
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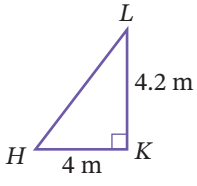
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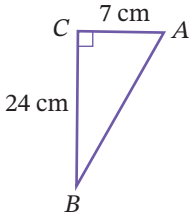
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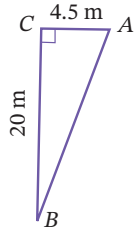
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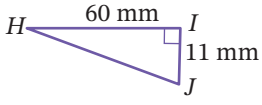
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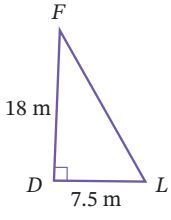
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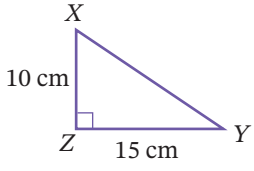
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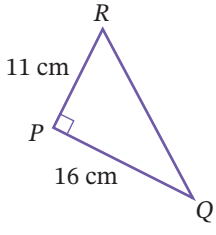
3 Find the length of the hypotenuse in each triangle, as a surd.

EXAMPLE 2

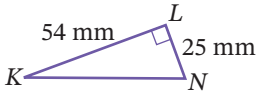
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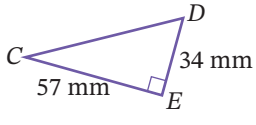
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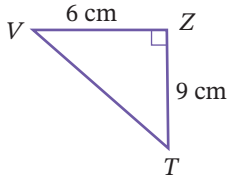
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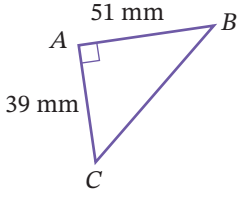
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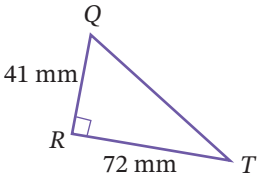
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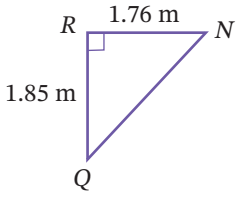
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4 Find the length of the hypotenuse in each triangle, correct to one decimal place.

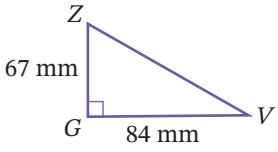
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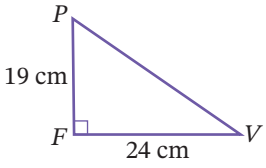
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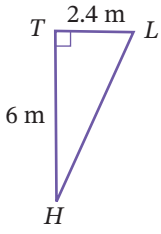
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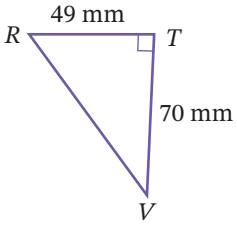
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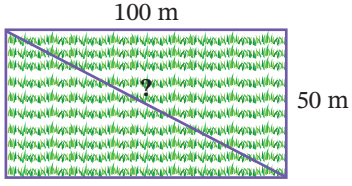
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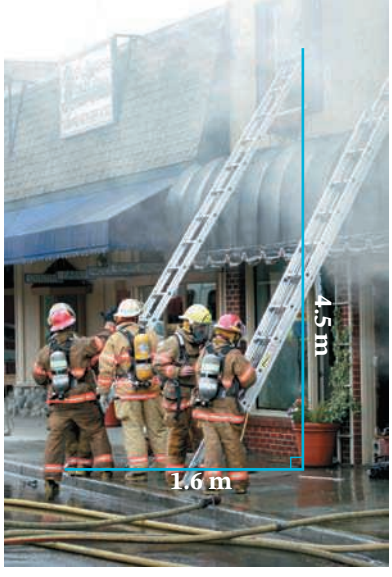
5 A rectangular field is 100 m long and 50 m wide. How far is it from one corner to the opposite corner, along the diagonal? Select the correct answer **A**, **B**, **C** or **D**.



A 150 m **B** 111.8 m

C 100.2 m **D** 98.3 m

6 A firefighter places a ladder on a window 4.5 m above the ground. If the foot of the ladder is 1.6 m from the wall, how long is the ladder, correct to one decimal place?



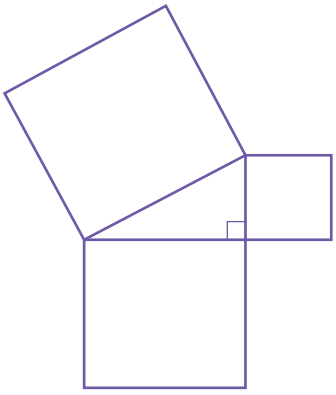
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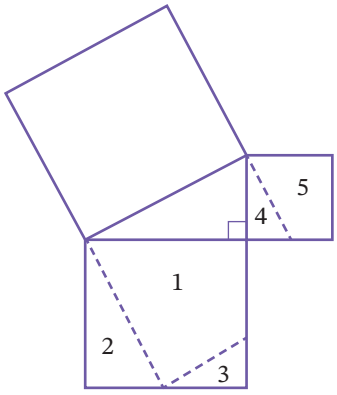
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7 Pythagoras' theorem can be demonstrated visually by dissection (cutting up shapes).
PS R

a On a sheet of paper, construct a right-angled triangle and draw squares on all 3 sides, as shown in the diagram on the left. Alternatively, download the worksheet 'Pythagoras 1'.





b In the 2 smaller squares, draw lines perpendicular to the hypotenuse as shown, and in the bottom square, draw a line parallel to the hypotenuse as shown in the diagram on the right.

c Number the regions 1, 2, 3, 4, 5 as shown.

d Cut out the 5 pieces and rearrange them to form the square on the hypotenuse.

e Does the area of the 2 smaller squares equal the area of the larger square?

f How does this 'prove' Pythagoras' theorem?

g Repeat this question on another right-angled triangle.

8 **a** In $\triangle ABC$, $\angle ABC = 90°$, $AB = 39$ cm and $BC = 57$ cm. Find AC, correct to one decimal place.

b In $\triangle MPQ$, $\angle PQM = 90°$, $QM = 2.4$ m and $PQ = 3.7$ m. Find PM as a surd.

c In $\triangle RVJ$, $\angle J = 90°$, $JV = 12.7$ cm and $JR = 42$ mm. Find RV, correct to the nearest millimetre.

d In $\triangle EGB$, $EG = EB = 127$ mm and $\angle GEB = 90°$. Find the length of BG, correct to one decimal place.

e In $\triangle VZX$, $\angle V = 90°$, $VX = 547$ cm and $VZ = 11.6$ m. Find ZX in metres, correct to the nearest 0.1 m.

f In $\triangle PQR$, $\angle RPQ = 90°$, $PQ = 2350$ mm and $PR = 5.8$ m. Find QR in metres, correct to 2 decimal places.

Worksheet
Pythagoras 1

2.01

EXAMPLE 3

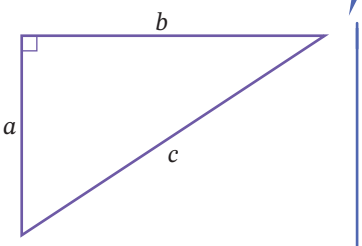
Foundation Standard Complex

Skillsheet
Spreadsheets

Technology

Finding the hypotenuse

Use a spreadsheet to calculate the length of the hypotenuse, c, of a right-angled triangle, given the lengths of the other 2 sides (a and b).



1 Enter the headings shown below into cells A1, B1, C1 and D1. Then enter the values shown into columns A and B.

	A	B	C	D	E
1	a	b	a^2+b^2	c	
2	3	4			
3	5	12			
4	6	8			
5	7	24			
6					
7					

2 Click on cell C2. Type the formula **=A2^2+B2^2**. Use **Fill Down** to copy this formula into cells C3 to C5.

3 To calculate each hypotenuse, we need to find the square root of the values in column C each time. So, in cell D2, enter **=sqrt(C2)**. Then click on cell D2 and use **Fill Down** to copy this formula into cells D3 to D5.

4 Use your spreadsheet to find the length of the hypotenuse, given the following pairs of values for the other 2 sides.

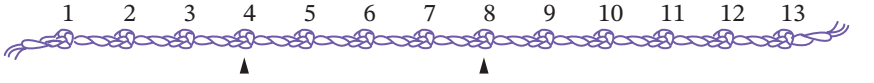
a $a = 12, b = 9$ **b** $a = 10, b = 10$ **c** $a = 7, b = 6$

d $a = 1, b = 8$ **e** $a = 7.5, b = 11.9$ **f** $a = 6.4, b = 2.3$

Did you know?

Egyptian rope stretchers

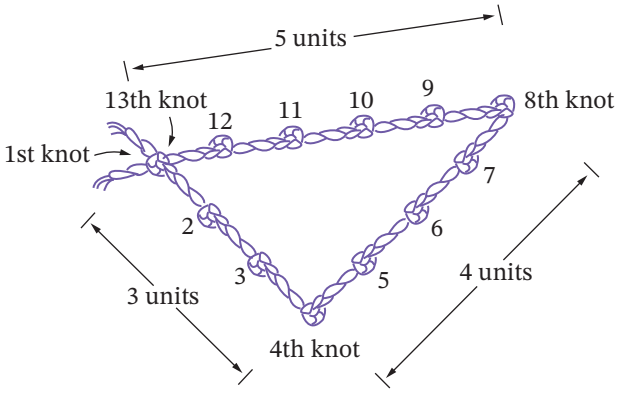
In ancient times, whenever the Nile River flooded, Egyptian farmers would have to rebuild fences around their fields afterwards. To form right angles for this, they used a secret 'magical' technique called rope stretching to create a 90° angle, making a length of rope with 13 knots tied in it at equal intervals.



The rope was fixed to the ground at the 4th and 8th knots, and stretched around to create a triangle with side lengths 3, 4 and 5 units. The right angle is formed at the 4th knot.

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The builders of the great pyramids also used rope stretching, but could not understand why this 'trick' produced a right angle. Pythagoras studied with the rope stretchers and eventually worked it out.

Make and stretch a knotted rope to create a right angle using a 3-4-5 triangle.

Surds 2.02

- $\sqrt{25} = 5$ because $5^2 = 25$ 'the square root of 25'
- $\sqrt{81} = 9$ because $9^2 = 81$ 'the square root of 81'

Most square roots do not give exact answers like the ones above.

For example, $\sqrt{7} = 2.645751311\ldots \approx 2.6$. Such roots are called **surds**.

A surd is a square root ($\sqrt{\ }$), cube root ($\sqrt[3]{\ }$), or any type of root whose exact decimal or fraction value cannot be found. As a decimal, its digits run endlessly *without repeating* (like π), so they are neither terminating nor recurring decimals.

Rational numbers such as fractions, decimals and percentages, can be expressed in the form $\frac{a}{b}$, where a and b are integers (and $b \neq 0$), but surds are **irrational numbers** because they cannot be expressed in this form. The rational and irrational numbers together make up the **real numbers**.

Example 4

Select the surds from this list of square roots:

$$\sqrt{72} \quad \sqrt{121} \quad \sqrt{64} \quad \sqrt{90} \quad \sqrt{28}$$

Solution

$\sqrt{72} = 8.4852...$

$\sqrt{72}$ is called an '*exact* value' while 8.4852 is called its '*approximate* value'

$\sqrt{121} = 11$

$\sqrt{64} = 8$ $\sqrt{90} = 9.4868...$ $\sqrt{28} = 5.2915...$

so the surds are $\sqrt{72}$, $\sqrt{90}$ and $\sqrt{28}$.

Video
Surds and irrational numbers

Video
Surds and irrational numbers

Example 5

Is each number rational or irrational?

a $4\frac{2}{5}$ **b** $\sqrt[3]{-8}$ **c** $-\sqrt{7}$ **d** $0.\dot{6}$ **e** 5π

Solution

a $4\frac{2}{5} = \frac{22}{5}$ which is in the form of a fraction $\frac{a}{b}$

$\therefore 4\frac{2}{5}$ is a rational number.

b $\sqrt[3]{-8} = -2$ which can be written as $\frac{-2}{1}$

$\therefore \sqrt[3]{-8}$ is a rational number.

c $-\sqrt{7} = -2.645\,751\,311\ldots$ The digits run endlessly without repeating

$\therefore -\sqrt{7}$ is an irrational number.

d $0.\dot{6} = 0.666\ldots$ which is a recurring decimal

$= \frac{2}{3}$ which is a fraction

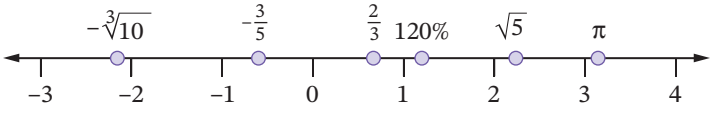
$\therefore 0.\dot{6}$ is a rational number.

e $5\pi = 15.707\,963\,27...$ The digits run endlessly without repeating

$\therefore 5\pi$ is an irrational number.

Surds on a number line

The rational and irrational numbers together make up the **real numbers**. Any real number can be represented by a point on the number line.



$-\sqrt[3]{10} = -2.1544\ldots$	irrational (surd)	$120\% = 1.2$	rational (percentage)
$-\frac{3}{5} = -0.6$	rational (fraction)	$\sqrt{5} = 2.2360\ldots$	irrational (surd)
$\frac{2}{3} = 0.6666\ldots$	rational (fraction)	$\pi = 3.1415...$	irrational (pi)

Example 6

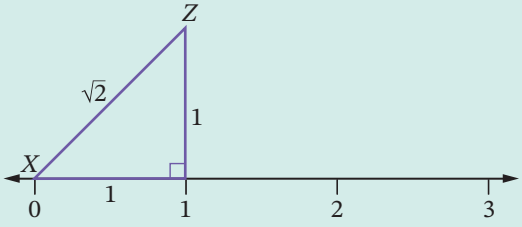
Use a pair of compasses and Pythagoras' theorem to estimate the value of $\sqrt{2}$ on a number line.

Solution

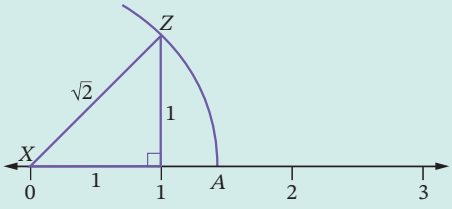
Step 1: Using a scale of 1 unit to 2 cm, draw a number line as shown.



Step 2: Construct a right-angled triangle on the number line with base length and height 1 unit as shown. By Pythagoras' theorem, show that $XZ = \sqrt{2}$ units.



Step 3: With 0 as centre, use compasses with radius XZ ($\sqrt{2}$) to draw an arc to meet the number line at A as shown. The point A represents the value of $\sqrt{2}$ and should be approximately 1.4142...



EXERCISE 2.02 ANSWERS ON P. 613

Surds **U F R C**

1 Which one of the following is a surd? Select the correct answer **A**, **B**, **C** or **D**. **C**

A $\sqrt{9}$ **B** $\sqrt{225}$ **C** $\sqrt{160}$ **D** $\sqrt{81}$

2 Which one of the following is NOT a surd? Select **A**, **B**, **C** or **D**. **C**

A $\sqrt{77}$ **B** $\sqrt{144}$ **C** $\sqrt{18}$ **D** $\sqrt{200}$

3 Select the surds from the following list of square roots. **C**

$\sqrt{32}$ $\sqrt{33}$ $\sqrt{289}$ $\sqrt{81}$ $\sqrt{4.9}$ $\sqrt{52}$ $\sqrt{121}$ $\sqrt{144}$ $\sqrt{196}$ $\sqrt{200}$

Worksheet
Surds on the number line

2.02

EXAMPLE 4

Foundation Standard Complex

EXAMPLE 5

4 Is each number rational (R) or irrational (I)? R C

a $5.\dot{6}$ **b** $\sqrt{8}$ **c** $\sqrt{4}$ **d** $3\frac{1}{7}$

e $\sqrt[3]{27}$ **f** $1.3\dot{5}$ **g** $\sqrt[3]{-64}$ **h** $27\frac{1}{2}\%$

i $\sqrt{5\times10^3}$ **j** $\frac{3}{11}$ **k** $\frac{\sqrt{50}}{3}$ **l** $\sqrt{\sqrt{4}}$

5 Arrange each set of numbers in descending order.

a $1\frac{4}{7}, \sqrt{2}, \frac{\pi}{2}$ **b** $\sqrt[3]{20}, 2.\dot{6}, 2\frac{7}{9}$

6 Use the method from Example **6** to estimate the value of $\sqrt{2}$ on a number line.

EXAMPLE 6

7 **a** Use the method from Example **6** to estimate the value of $\sqrt{5}$ on a number line by constructing a right-angled triangle with base length 2 units and height 1 unit.

b Use a similar method to estimate the following surds on a number line.

i $\sqrt{10}$ **ii** $\sqrt{17}$

8 Without using a calculator, determine between which 2 consecutive numbers the value of $\sqrt{40}$ lies. Select **A**, **B**, **C** or **D**. R

A 4 and 5 **B** 5 and 6 **C** 6 and 7 **D** 7 and 8

9 Is the statement $\sqrt{x^2} = x$ always true? R

10 Is the statement $\left(\sqrt{x}\right)^2 = x$ always true? R

Investigation

Proof that $\sqrt{2}$ is irrational

A method of proof sometimes used in mathematics is to assume the **opposite** of what is being proved, and to show that it is **impossible**. This is called a **proof by contradiction**, and we will use it to prove that $\sqrt{2}$ is irrational.

First, we assume that $\sqrt{2}$ is rational. So, assume that $\sqrt{2}$ can be written as a simplified fraction $\frac{a}{b}$, where a and b are integers ($b \neq 0$) with no common factors.

$\sqrt{2} = \frac{a}{b}$

$2 = \frac{a^2}{b^2}$ Squaring both sides

$a^2 = 2b^2$

$2b^2$ is an even number because it is divisible by 2, $\therefore a^2$ is even.

$\therefore a$ is even, because an even integer multiplied by itself is always even and an odd integer multiplied by itself is always odd.

$\therefore a = 2m$, where m is another integer.

$\therefore a^2 = (2m)^2 = 2b^2$

$4m^2 = 2b^2$

$2m^2 = b^2$

Foundation Standard Complex

$b^2 = 2m^2$

$\therefore b^2$ is even

$\therefore b$ is even.

But a and b can't **both** be even because this contradicts the assumption that a and b have no common factors. Therefore, the assumption that $\sqrt{2}$ is rational is false, so $\sqrt{2}$ must be irrational.

1 Use proof by contradiction to show that these surds are irrational:

a $\sqrt{3}$ **b** $\sqrt{5}$

2 Compare your proofs with those of other students.

Simplifying surds 2.03

The square of any real number is always positive (except for $0^2 = 0$), so it is not possible to give the square root of a *negative* number.

The radical symbol ($\sqrt{\ }$) stands for the *positive* square root of a number, for example $\sqrt{4} = 2$ (not -2).

The square root of x

For $x < 0$ (negative), $\sqrt{x}$ is undefined.

For $x = 0$, $\sqrt{x} = 0$.

For $x > 0$ (positive), $\sqrt{x}$ is the positive square root of x.

For $x \geq 0$, $\left(\sqrt{x}\right)^2 = \sqrt{x} \times \sqrt{x} = x$ and $\sqrt{x^2} = x$.

Example 7

Simplify each expression.

a $\left(\sqrt{7}\right)^2$ **b** $\left(3\sqrt{5}\right)^2$ **c** $\left(-2\sqrt{3}\right)^2$

Solution

a $\left(\sqrt{7}\right)^2 = 7$

b $(3\sqrt{5})^2 = 3\sqrt{5} \times 3\sqrt{5}$ $3\sqrt{5}$ means $3 \times \sqrt{5}$

$= 3^2 \times (\sqrt{5})^2$

$= 9 \times 5$

$= 45$

c $(-2\sqrt{3})^2 = (-2)^2 \times (\sqrt{3})^2$

$= 4 \times 3$

$= 12$

PATH



Worksheet Surds

Puzzle Simplifying surds

Technology Simplifying surds quiz

Spreadsheet Simplifying surds quiz

PATH

The square root of a product

For $x > 0$ and $y > 0$:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

A surd $\sqrt{n}$ can be simplified if n can be divided into 2 factors, where one of them is a square number such as 4, 9, 16, 25, 36, 49, ...

Videos
Simplifying surds 1
Simplifying surds 2

Example 8

Simplify each surd.

a $\sqrt{50}$ **b** $\sqrt{432}$ **c** $4\sqrt{12}$ **d** $\frac{\sqrt{288}}{3}$

Solution

a $\sqrt{50} = \sqrt{25} \times \sqrt{2}$ 25 is a square number
$= 5 \times \sqrt{2}$
$= 5\sqrt{2}$

b **Method 1**

$\sqrt{432} = \sqrt{4} \times \sqrt{108}$
$= 2 \times \sqrt{108}$
$= 2 \times \sqrt{36} \times \sqrt{3}$
$= 2 \times 6 \times \sqrt{3}$
$= 12 \times \sqrt{3}$
$= 12\sqrt{3}$

Method 2

$\sqrt{432} = \sqrt{144} \times \sqrt{3}$
$= 12 \times \sqrt{3}$
$= 12\sqrt{3}$

Method 1 involves simplifying surds *twice* ($\sqrt{432}$ and $\sqrt{108}$). Method 2 shows that when simplifying surds, it is more efficient to first look for the highest square factor possible.

c $4\sqrt{12} = 4 \times \sqrt{4} \times \sqrt{3}$
$= 4 \times 2 \times \sqrt{3}$
$= 8\sqrt{3}$

d $\frac{\sqrt{288}}{3} = \frac{\sqrt{144} \times \sqrt{2}}{3}$
$= \frac{12\sqrt{2}}{3}$
$= \frac{12^{4}\sqrt{2}}{3_{1}}$
$= 4\sqrt{2}$

EXERCISE 2.03 ANSWERS ON P. 613

Simplifying surds U F R C

EXAMPLE 7

1 Simplify each expression.

a $\left(\sqrt{2}\right)^2$ **b** $\left(\sqrt{5}\right)^2$ **c** $\left(3\sqrt{3}\right)^2$ **d** $\left(5\sqrt{10}\right)^2$

e $\left(\sqrt{0.09}\right)^2$ **f** $\left(-2\sqrt{7}\right)^2$ **g** $\left(-3\sqrt{5}\right)^2$ **h** $\left(-5\sqrt{2}\right)^2$

Foundation Standard Complex

2 Simplify each surd.

a $\sqrt{8}$	**b** $\sqrt{27}$	**c** $\sqrt{24}$	**d** $\sqrt{54}$
e $\sqrt{243}$	**f** $\sqrt{45}$	**g** $\sqrt{48}$	**h** $\sqrt{200}$
i $\sqrt{96}$	**j** $\sqrt{63}$	**k** $\sqrt{288}$	**l** $\sqrt{108}$
m $\sqrt{75}$	**n** $\sqrt{147}$	**o** $\sqrt{32}$	**p** $\sqrt{242}$
q $\sqrt{162}$	**r** $\sqrt{245}$	**s** $\sqrt{125}$	**t** $\sqrt{512}$

3 Simplify each expression.

a $5\sqrt{50}$	**b** $3\sqrt{8}$	**c** $4\sqrt{27}$	**d** $8\sqrt{98}$
e $\frac{\sqrt{40}}{2}$	**f** $\frac{\sqrt{243}}{9}$	**g** $\frac{\sqrt{28}}{6}$	**h** $3\sqrt{24}$
i $9\sqrt{68}$	**j** $\frac{\sqrt{3125}}{10}$	**k** $\frac{1}{2}\sqrt{72}$	**l** $\frac{3}{4}\sqrt{48}$
m $10\sqrt{160}$	**n** $3\sqrt{75}$	**o** $7\sqrt{68}$	**p** $\frac{\sqrt{52}}{6}$

4 Decide whether each statement is true (T) or false (F). R

a $3\sqrt{7} = \sqrt{21}$	**b** $\sqrt{12} = 6$
c $\left(\sqrt{9.4}\right)^2 = 9.4$	**d** $\sqrt{75} = 5\sqrt{3}$
e $\sqrt{3} \approx 1.7$	**f** The exact value of $\sqrt{10}$ is 3.162 277 8

5 What is $\sqrt{x^2}$ if $x < 0$? R C

PATH

EXAMPLE 8

2.03

Mental skills 2A: Maths without calculators ANSWERS ON P. 614

Quiz
Mental skills 2A

Multiplying and dividing by a power of 10

Multiplying a number by 10, 100, 1000, etc. moves the decimal point to the right and makes the number bigger. We place 0s at the end of the number if necessary.

- When multiplying a number by **10**, move the decimal point **1** place to the right.
- When multiplying a number by **100**, move the decimal point **2** places to the right.
- When multiplying a number by **1000**, move the decimal point **3** places to the right.

The number of places the decimal is moved to the right matches the number of 0s in the 10, 100 or 1000 we are multiplying by.

1 Study each example.

a	$26.32 \times 10 = 26.32 = 263.2$	The point moves 1 place to the right.
b	$8.701 \times 100 = 8.70\,1 = 870.1$	The point moves 2 places to the right.
c	$6.01 \times 1000 = 6.010 = 6010$	The point moves 3 places to the right after a 0 is placed at the end.
d	$17 \times 100 = 17.00 = 1700$	The point moves 2 places to the right after 2 0s are placed at the end.

Foundation Standard Complex

2 Now evaluate each expression.

a	89.54×10	**b**	3.7×10	**c**	0.831×100
d	42×100	**e**	5.2716×1000	**f**	156.1×10
g	31.84×1000	**h**	64.3×100	**i**	0.0224×1000
j	4.894×10	**k**	7.389×1000	**l**	11.42×100

Dividing a number by 10, 100, 1000, etc. moves the decimal point to the left and makes the number smaller. We place 0s at the start of the decimal if necessary.

- When dividing a number by **10**, move the decimal point **1** place to the left.
- When dividing a number by **100**, move the decimal point **2** places to the left.
- When dividing a number by **1000**, move the decimal point **3** places to the left.

3 Study each example.

a	$145.66 \div 10 = 145.66 = 14.566$	The point moves 1 place to the left.
b	$2.357 \div 100 = 002.357 = 0.023\,57$	The point moves 2 places to the left after 2 0s are inserted at the start.
c	$14.9 \div 1000 = 0014.9 = 0.0149$	The point moves 3 places to the left after 2 0s are inserted at the start.
d	$45 \div 100 = 045. = 0.45$	The point moves 2 places to the left after 1 zero is inserted at the start.

4 Now evaluate each expression.

a	$733.4 \div 10$	**b**	$9.4 \div 10$	**c**	$652 \div 100$	**d**	$10.4 \div 100$
e	$704 \div 1000$	**f**	$198.5 \div 100$	**g**	$2 \div 100$	**h**	$4159 \div 1000$
i	$123 \div 10$	**j**	$0.758 \div 100$	**k**	$8.49 \div 100$	**l**	$25.1 \div 1000$

Investigation

A formula for calculating square roots

Calculators and computers use a formula repeatedly to give an approximate decimal answer for the square root. The formula for calculating $\sqrt{M}$ is $x_{n+1} = \frac{1}{2}\left(x_n + \frac{M}{x_n}\right)$, where x_0 is the first guess and each calculation of the formula gives a better approximation than the last one.

For example, to calculate $\sqrt{500}$ using a first guess of $x_0 = 20$:

$$x_1 = \frac{1}{2}\left(20 + \frac{500}{20}\right) = 22.5$$

$$x_2 = \frac{1}{2}\left(22.5 + \frac{500}{22.5}\right) = 22.361\,111\,11$$

$$x_3 = \frac{1}{2}\left(22.36 + \frac{500}{22.36}\right) = 22.360\,679\,78$$

This process can continue endlessly, with the accuracy increasing each time. This is called an **iterative** formula or process, where iterative means 'repeated'.

$\therefore \sqrt{500} \approx 22.36$, correct to 2 decimal places.

1 **a** Estimate $\sqrt{55}$.

b Use the formula to evaluate $\sqrt{55}$ to 3 decimal places, using the iterative formula.

c Evaluate $\sqrt{55}$ on your calculator to check your answer.

2 **a** Estimate $\sqrt{700}$.

b Use the formula to evaluate $\sqrt{700}$ to 4 decimal places.

c Evaluate $\sqrt{700}$ on your calculator to check your answer.

3 Design a spreadsheet that uses the formula repeatedly to calculate square roots.

Adding and subtracting surds

2.04

Just as you can only add or subtract 'like terms' in algebra, you can only add or subtract 'like surds'.

PATH



Video Adding and subtracting surds



Puzzle Surds code puzzle

Example 9

Simplify each expression.

a $4\sqrt{2}+5\sqrt{2}$ **b** $7\sqrt{3}-2\sqrt{3}$ **c** $5\sqrt{2}-3\sqrt{3}+\sqrt{2}$

d $\sqrt{50}+\sqrt{32}$ **e** $\sqrt{8}-\sqrt{27}+\sqrt{18}$ **f** $5\sqrt{20}-3\sqrt{125}$

Solution

a $4\sqrt{2}+5\sqrt{2}=9\sqrt{2}$

b $7\sqrt{3}-2\sqrt{3}=5\sqrt{3}$

c $5\sqrt{2}-3\sqrt{3}+\sqrt{2}=6\sqrt{2}-3\sqrt{3}$

d $\sqrt{50}+\sqrt{32}=\sqrt{25}\sqrt{2}+\sqrt{16}\sqrt{2}$
$=5\sqrt{2}+4\sqrt{2}$
$=9\sqrt{2}$

Simplifying each surd

e $\sqrt{8}-\sqrt{27}+\sqrt{18}=\sqrt{4}\sqrt{2}-\sqrt{9}\sqrt{3}+\sqrt{9}\sqrt{2}$
$=2\sqrt{2}-3\sqrt{3}+3\sqrt{2}$
$=5\sqrt{2}-3\sqrt{3}$

f $5\sqrt{20}-3\sqrt{125}=5\sqrt{4}\sqrt{5}-3\sqrt{25}\sqrt{5}$
$=5\times 2\sqrt{5}-3\times 5\sqrt{5}$
$=10\sqrt{5}-15\sqrt{5}$
$=-5\sqrt{5}$

PATH

EXERCISE 2.04 ANSWERS ON P. 614

Adding and subtracting surds U F

EXAMPLE 9

1 Simplify each vexpression.

a $5\sqrt{7}+2\sqrt{7}$ **b** $3\sqrt{2}-8\sqrt{2}$

c $7\sqrt{5}-\sqrt{5}$ **d** $\sqrt{5}+3\sqrt{5}$

e $5\sqrt{17}-5\sqrt{17}$ **f** $3\sqrt{10}-2\sqrt{10}$

g $4\sqrt{15}-3\sqrt{15}+7\sqrt{15}$ **h** $5\sqrt{6}-2\sqrt{6}-4\sqrt{6}$

i $3\sqrt{3}+4\sqrt{3}-5\sqrt{3}$ **j** $4\sqrt{5}+7\sqrt{5}-\sqrt{5}$

k $8\sqrt{10}-5\sqrt{10}+3\sqrt{10}$ **l** $10\sqrt{3}-3\sqrt{3}-12\sqrt{3}$

2 Simplify each expression.

a $3\sqrt{5}-8+2\sqrt{5}$ **b** $11\sqrt{10}+3\sqrt{2}+2\sqrt{10}$

c $-4\sqrt{3}+5\sqrt{2}-5\sqrt{3}$ **d** $3\sqrt{15}+3\sqrt{2}+4\sqrt{15}+5\sqrt{2}$

e $\sqrt{7}-3\sqrt{5}-4\sqrt{7}+\sqrt{5}$ **f** $4\sqrt{6}-3\sqrt{3}-2\sqrt{6}-5\sqrt{3}$

g $10\sqrt{11}-5\sqrt{3}+3\sqrt{11}+4\sqrt{3}$ **h** $\sqrt{13}+8\sqrt{7}-7\sqrt{13}+3\sqrt{7}$

i $2\sqrt{5}-3\sqrt{7}-2\sqrt{5}-3\sqrt{7}$ **j** $4\sqrt{10}-3\sqrt{5}-4\sqrt{10}$

3 For each expression, select the correct simplified answer **A**, **B**, **C** or **D**.

a $\sqrt{3}+\sqrt{12}$

A $5\sqrt{3}$ **B** $\sqrt{15}$ **C** $2\sqrt{6}$ **D** $3\sqrt{3}$

b $4\sqrt{5}-2\sqrt{125}$

A $-6\sqrt{5}$ **B** $\sqrt{5}$ **C** $-\sqrt{45}$ **D** $-46\sqrt{5}$

4 Simplify each expression.

a $\sqrt{8}+\sqrt{32}$ **b** $\sqrt{108}-\sqrt{27}$ **c** $\sqrt{20}-\sqrt{80}$

d $\sqrt{28}-\sqrt{63}$ **e** $3\sqrt{6}+\sqrt{24}$ **f** $2\sqrt{5}+\sqrt{125}$

g $\sqrt{40}-\sqrt{90}$ **h** $5\sqrt{11}+\sqrt{99}$ **i** $3\sqrt{2}+\sqrt{18}$

j $\sqrt{27}+5\sqrt{3}$ **k** $\sqrt{200}-7\sqrt{2}$ **l** $\sqrt{50}+\sqrt{32}$

m $5\sqrt{3}+2\sqrt{27}$ **n** $3\sqrt{20}-\sqrt{245}$ **o** $7\sqrt{12}-5\sqrt{48}$

p $4\sqrt{27}+2\sqrt{243}$ **q** $3\sqrt{63}-2\sqrt{28}$ **r** $2\sqrt{98}+3\sqrt{162}$

s $-5\sqrt{6}+2\sqrt{150}$ **t** $4\sqrt{50}+3\sqrt{18}$ **u** $5\sqrt{27}-6\sqrt{75}$

v $3\sqrt{112}-2\sqrt{252}$ **w** $\sqrt{32}+\sqrt{8}+\sqrt{12}$ **x** $\sqrt{27}+\sqrt{54}+\sqrt{243}$

y $\sqrt{98}-3\sqrt{20}-2\sqrt{8}$ **z** $3\sqrt{96}-2\sqrt{150}+\sqrt{24}$

Foundation Standard Complex

9780170479585

Multiplying and dividing surds

2.05

PATH

The square root of products and quotients

For $x > 0$ and $y > 0$:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

$$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$$

Example 10

Simplify each expression.

a $\sqrt{3} \times \sqrt{5}$ **b** $\sqrt{10} \times \sqrt{6}$ **c** $3\sqrt{7} \times 5\sqrt{7}$

d $5\sqrt{27} \times 3\sqrt{6}$ **e** $\sqrt{54} \div (-\sqrt{2})$ **f** $\frac{15\sqrt{32}}{5\sqrt{8}}$

Solution

a $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

b $\sqrt{10} \times \sqrt{6} = \sqrt{60}$
$= \sqrt{4} \times \sqrt{15}$
$= 2\sqrt{15}$

c $3\sqrt{7} \times 5\sqrt{7} = 3 \times 5 \times \sqrt{7} \times \sqrt{7}$
$= 15 \times 7$
$= 105$

d $5\sqrt{27} \times 3\sqrt{6} = 5 \times 3 \times \sqrt{27} \times \sqrt{6}$
$= 15\sqrt{162}$
$= 15 \times \sqrt{81} \times \sqrt{2}$
$= 15 \times 9\sqrt{2}$
$= 135\sqrt{2}$

e $\sqrt{54} \div (-\sqrt{2}) = -\frac{\sqrt{54}}{\sqrt{2}}$
$= -\sqrt{27}$
$= -\sqrt{9} \times \sqrt{3}$
$= -3\sqrt{3}$

f $\frac{15\sqrt{32}}{5\sqrt{8}} = 3\sqrt{4}$
$= 3 \times 2$
$= 6$

Example 11

Simplify $\frac{5\sqrt{2} \times 4\sqrt{12}}{10\sqrt{8}}$.

Solution

$\frac{5\sqrt{2} \times 4\sqrt{12}}{10\sqrt{8}} = \frac{20\sqrt{24}}{10\sqrt{8}}$
$= 2\sqrt{3}$

PATH

EXERCISE 2.05 ANSWERS ON P. 614

Multiplying and dividing surds U F

EXAMPLE 10

1 Simplify each expression.

a $\sqrt{7} \times \sqrt{2}$ **b** $-\sqrt{5} \times \sqrt{7}$ **c** $\sqrt{6} \times \sqrt{8}$

d $\sqrt{12} \times \sqrt{3}$ **e** $\sqrt{10} \times \left(-\sqrt{5}\right)$ **f** $3\sqrt{3} \times 5\sqrt{3}$

g $5\sqrt{10} \times 3\sqrt{3}$ **h** $-2\sqrt{7} \times 5\sqrt{3}$ **i** $7\sqrt{5} \times 4\sqrt{5}$

j $2\sqrt{3} \times \left(-5\sqrt{6}\right)$ **k** $4\sqrt{3} \times \sqrt{27}$ **l** $-3\sqrt{5} \times 4\sqrt{10}$

m $-7\sqrt{2} \times 4\sqrt{8}$ **n** $\sqrt{18} \times 8\sqrt{3}$ **o** $10\sqrt{2} \times 2\sqrt{8}$

p $3\sqrt{18} \times 5\sqrt{12}$ **q** $3\sqrt{44} \times \left(-2\sqrt{99}\right)$ **r** $5\sqrt{8} \times 4\sqrt{40}$

s $8\sqrt{3} \times 3\sqrt{54}$ **t** $-8\sqrt{32} \times \sqrt{27}$ **u** $\sqrt{90} \times \sqrt{72}$

v $-5\sqrt{20} \times 3\sqrt{8}$ **w** $7\sqrt{18} \times 3\sqrt{24}$ **x** $3\sqrt{48} \times 2\sqrt{42}$

2 Simplify each expression.

a $\sqrt{15} \div \sqrt{3}$ **b** $\sqrt{18} \div \left(-\sqrt{6}\right)$ **c** $\dfrac{6\sqrt{48}}{2\sqrt{8}}$

d $10\sqrt{54} \div 5\sqrt{27}$ **e** $-3\sqrt{98} \div 6\sqrt{14}$ **f** $\dfrac{7\sqrt{18}}{\sqrt{2}}$

g $2\sqrt{24} \div 4\sqrt{6}$ **h** $\dfrac{\sqrt{128}}{\sqrt{2}}$ **i** $15\sqrt{18} \div 3\sqrt{6}$

j $\dfrac{-20\sqrt{10}}{-4\sqrt{5}}$ **k** $36\sqrt{24} \div 9\sqrt{8}$ **l** $16\sqrt{30} \div 8\sqrt{5}$

m $12\sqrt{14} \div 6$ **n** $\dfrac{3\sqrt{2}}{-12}$ **o** $\sqrt{80} \div 4\sqrt{5}$

p $5\sqrt{60} \div \sqrt{15}$ **q** $6\sqrt{8} \div 3\sqrt{2}$ **r** $\dfrac{-42\sqrt{54}}{6\sqrt{3}}$

s $12\sqrt{63} \div 3\sqrt{7}$ **t** $\dfrac{8\sqrt{50}}{2\sqrt{200}}$ **u** $6\sqrt{3} \div \sqrt{243}$

EXAMPLE 11

3 Simplify each expression.

a $\dfrac{3\sqrt{5} \times 4\sqrt{2}}{3\sqrt{40}}$ **b** $\dfrac{3\sqrt{12} \times 8\sqrt{6}}{4\sqrt{27}}$ **c** $\dfrac{5\sqrt{8} \times 2\sqrt{90}}{10\sqrt{24}}$

d $\dfrac{4\sqrt{5}}{2\sqrt{15} \times 5\sqrt{27}}$ **e** $\dfrac{10\sqrt{686} \times 3\sqrt{12}}{5\sqrt{28} \times \sqrt{18}}$ **f** $\dfrac{8\sqrt{80} \times 3\sqrt{2}}{4\sqrt{5} \times 6\sqrt{8}}$

Foundation Standard Complex

9780170479585

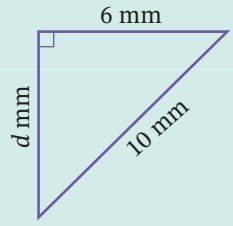
Finding a shorter side

2.06

Pythagoras' theorem can also be used to find the length of a shorter side of a right-angled triangle, if the hypotenuse and the other side are known.

Example 12

Find the value of d in this triangle.



Solution

We want to find the length of a shorter side.

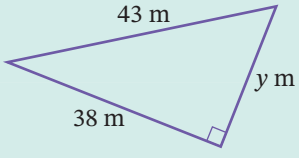
Using Pythagoras' theorem:

$10^2 = d^2 + 6^2$
$100 = d^2 + 36$
$d^2 + 36 = 100$
$d^2 = 100 - 36$
$= 64$
$d = \sqrt{64}$
$= 8$

From the diagram, a length of 8 mm looks reasonable because it must be shorter than the hypotenuse, which is 10 mm.

Example 13

Find the value of y as a surd for this triangle.



Solution

$43^2 = y^2 + 38^2$
$1849 = y^2 + 1444$
$y^2 + 1444 = 1849$
$y^2 = 1849 - 1444$
$= 405$
$y = \sqrt{405}$
$= 9\sqrt{5}$ Leave the answer as a surd

Worksheet
Finding an unknown side

Technology
Pythagoras' theorem

Spreadsheet
Pythagoras' theorem

Video
Pythagoras' theorem 2

Video
Pythagoras' theorem 1

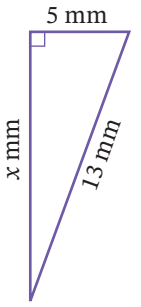
EXERCISE 2.06 ANSWERS ON P. 614

Finding a shorter side U F PS R

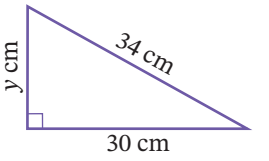
EXAMPLE 12

1 Find the value of the variable in each triangle.

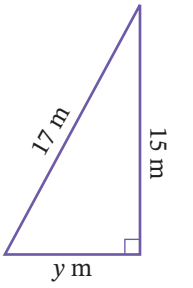
a



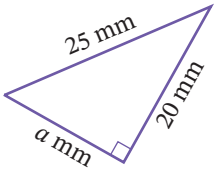
b



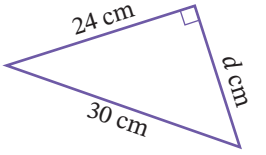
c



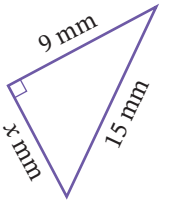
d



e

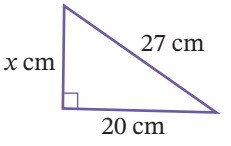


f

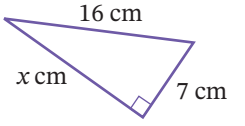


2 Find the value of the variable in each triangle, correct to one decimal place.

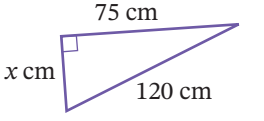
a



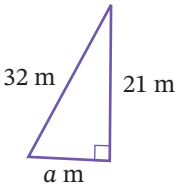
b



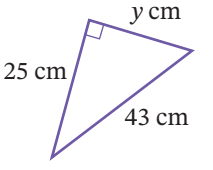
c



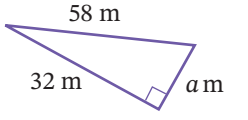
d



e



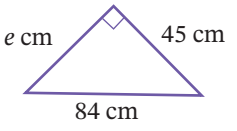
f



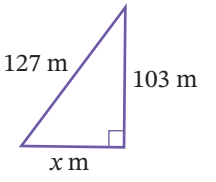
EXAMPLE 13

3 Find the value of the variable in each triangle as a surd.

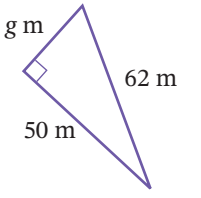
a



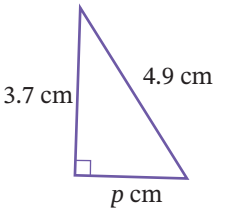
b



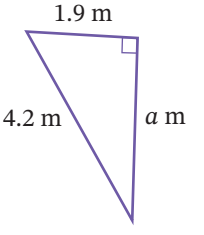
c



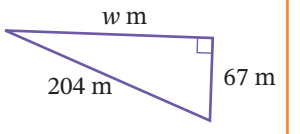
d



e



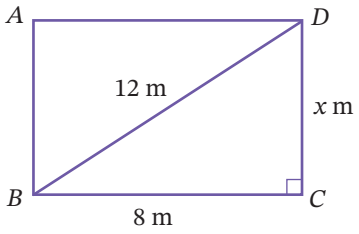
f



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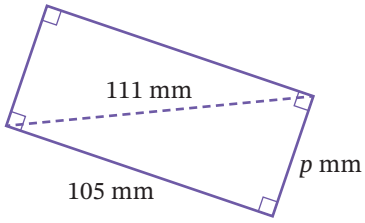
9780170479585

4 Find the value of x in this rectangle.
Select the correct answer **A**, **B**, **C** or **D**.



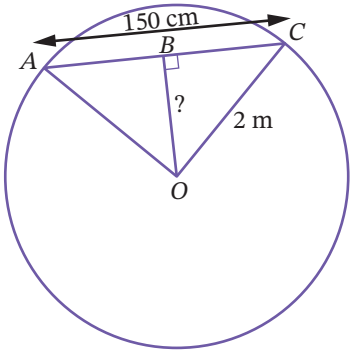
A $\sqrt{20}$
B $\sqrt{80}$
C $\sqrt{136}$
D $\sqrt{208}$

5 Find the value of p in this rectangle.
Select **A**, **B**, **C** or **D**.



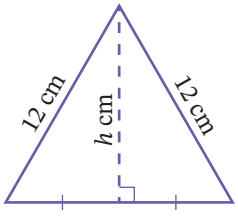
A 40
B 36
C 32
D 28

6 In this diagram, O is the centre of a circle.
A perpendicular line is drawn from O to B such that $OB \perp AC$ and $AB = BC$. Calculate the length of OB.
Select **A**, **B**, **C** or **D**. PS R



A 1.3 m
B 1.8 m
C 1.9 m
D 3.4 m

7 An equilateral triangle has sides of length 12 cm.
Find the perpendicular height, h cm, of the triangle, correct to 2 decimal places. PS R



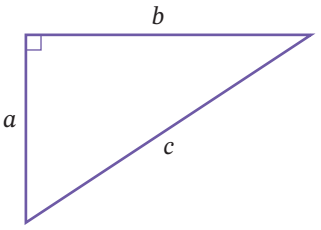
8 A square has a diagonal of length 30 cm. What is the length of each side of the square, correct to the nearest millimetre? PS R

Foundation Standard Complex

Technology

Finding a shorter side

Use a spreadsheet to calculate the length of a shorter side, b, of a right-angled triangle, given the length of the hypotenuse and the other side (c and a).



1 Enter the headings shown below into cells A1, B1, C1 and D1. Then enter the values shown into columns A and B.

	A	B	C	D
1	c	a	c^2 - a^2	b
2	25	20		
3	37	12		
4	18	16		
5	11	7		
6				

2 Click on cell C2. Type the formula **=A2^2-B2^2**. Use **Fill Down** to copy this formula into cells C3 to C5.

3 To calculate the values of b, we need to find the square root of each of the values in column C. So, in cell D2, enter **=sqrt(C2)**. Then click on cell D2 and use **Fill Down** to copy this formula into cells D3 to D5.

4 Use your spreadsheet to find the length of the unknown shorter side, given the following pairs of values for the hypotenuse and other side.

a $c = 61, a = 11$ **b** $c = 87, a = 60$ **c** $c = 35, a = 22$

d $c = 9.2, a = 2.7$ **e** $c = 158, a = 63$ **f** $c = 24.5, a = 12.8$

Pythagoras' theorem problems

2.07

When using Pythagoras' theorem to solve problems, it is useful to follow these steps.

- Read the problem carefully
- Draw a diagram involving a right-angled triangle and label any given information
- Choose a variable to represent the length or distance you want to find
- Use Pythagoras' theorem to find the value of the variable
- Answer the question

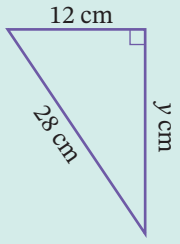
Video
Pythagoras' theorem problems

Worksheet
Applications of Pythagoras' theorem

Puzzles
Pythagoras 1
Pythagoras 2
Pythagorean two-step problems

Example 14

Find the value of y as a surd.



Solution

y is the length of a shorter side.

$$28^2 = y^2 + 12^2$$
$$784 = y^2 + 144$$
$$y^2 + 144 = 784$$
$$y^2 = 784 - 144$$
$$= 640$$
$$y = \sqrt{640}$$
$$= \sqrt{64} \times \sqrt{10}$$
$$= 8\sqrt{10}$$

Example 15

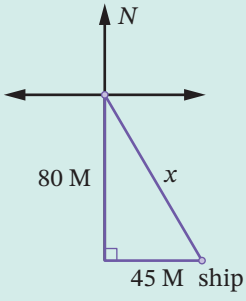
A ship sails 80 nautical miles south and then 45 nautical miles east. How far is it from its starting point, correct to one decimal place?

Solution

Let x be the distance the ship is from the starting point.

$$x^2 = 80^2 + 45^2$$
$$= 8425$$
$$x = \sqrt{8425}$$
$$= 91.7877\ldots$$
$$\approx 91.8\text{ M}$$

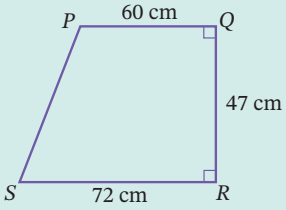
From the diagram, this looks like a reasonable answer



M is the symbol for nautical miles

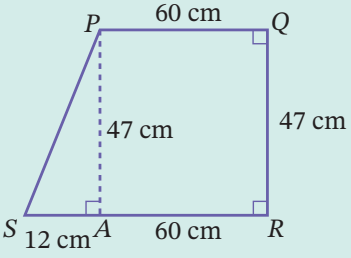
Example 16

Find the perimeter of this trapezium, correct to one decimal place.



Solution

The unknown side is PS. Form the right-angled $\triangle PSA$, as shown.



By comparing sides:

$PA = QR = 47$ cm and $AR = PQ = 60$ cm.

$\therefore SA = 72 - 60 = 12$ cm

$\therefore PS^2 = 47^2 + 12^2$

$= 2353$

$PS = \sqrt{2353}$

$= 48.5077...$

Perimeter $= 60 + 48.5077... + 72 + 47$

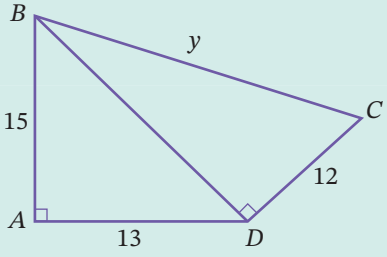
$= 227.5077...$

≈ 227.5 cm

From the diagram, this looks like a reasonable answer.

Example 17

Find the value of y, correct to 2 decimal places.



Solution

We need to find BD first.

In $\triangle ABD$,

$BD^2 = 15^2 + 13^2$

$= 394$

$BD = \sqrt{394}$ Leave BD as a surd for further working

In $\triangle BCD$,

$y^2 = \left(\sqrt{394}\right)^2 + 12^2$ $\left(\sqrt{394}\right)^2 = 394$

$= 538$

$y = \sqrt{538}$

$= 23.1948...$

≈ 23.19

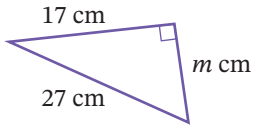
From the diagram, this looks like a reasonable answer.

EXERCISE 2.07 ANSWERS ON P. 614

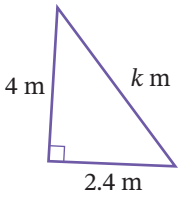
Pythagoras' theorem problems U F PS R

1 Find the value of the variable in each triangle, correct to one decimal place.

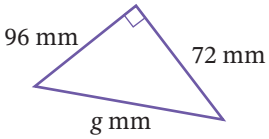
a



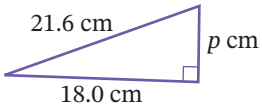
b



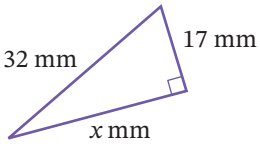
c



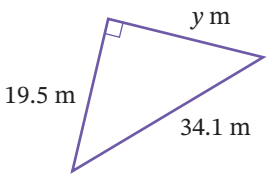
d



e



f

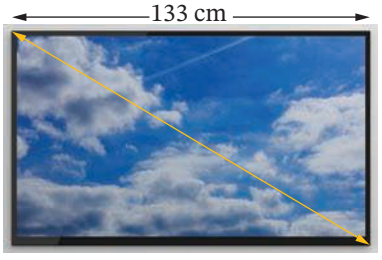


2 A ladder 5 m long is leaning against a wall. If the base of the ladder is 2 m from the bottom of the wall, how far does the ladder reach up the wall? Answer correct to 2 decimal places. PS



iStock.com/bmcent1

3 The size of a TV screen is described by the length of its diagonal. If a TV screen is 133 cm wide and 75 cm high, what is the size of the screen? Answer to the nearest centimetre.



Shutterstock.com/rawf8

4 Cooper wanted to find the length XY of the lake shown. He placed a marker at Z so that $\angle YXZ = 90°$. He measured XZ to be 450 m long and ZY to be 780 m long. What is the length of the lake? Answer correct to one decimal place.



Shutterstock.com/.Jamen Percy

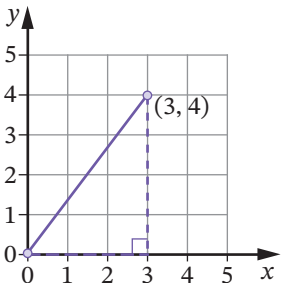
Foundation | Standard | Complex

EXAMPLE 15

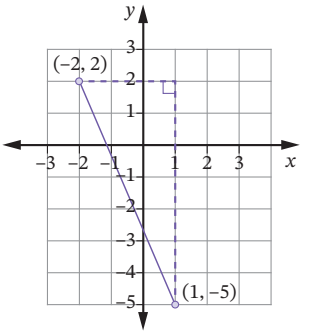
5 A ship sails 70 nautical miles west and then 60 nautical miles north. How far is it from its starting point, correct to one decimal place? PS

6 Use Pythagoras' theorem to calculate the distance between the 2 points on each number plane. In part **b**, write the answer as a surd. R

a



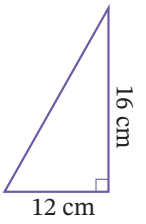
b



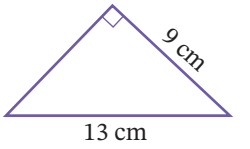
EXAMPLE 16

7 Calculate the perimeter of each shape, correct to one decimal place where necessary.

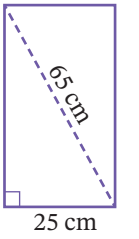
a



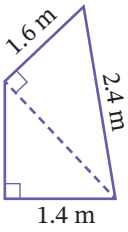
b



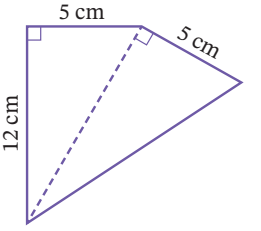
c



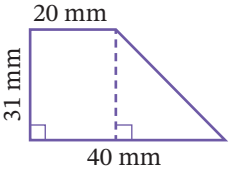
d



e

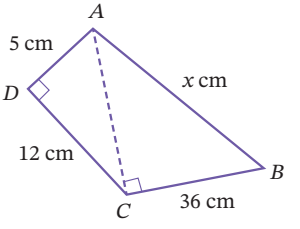


f

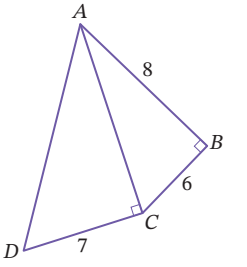


EXAMPLE 17

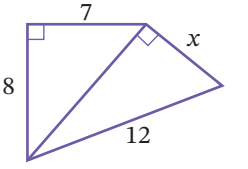
8 **a** Find the value of x, correct to 1 decimal place. R



b Find the length of AD, correct to 2 decimal places. R



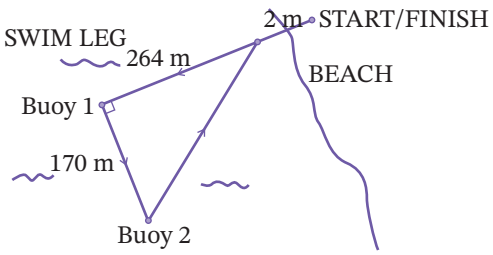
c Calculate the length of x as a surd. R



Foundation Standard Complex

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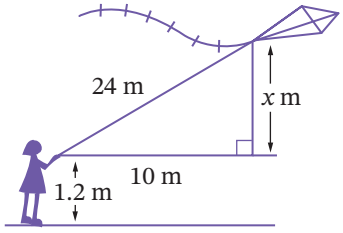
9 The swim course of a triathlon race has the shape of a right-angled triangle joined by a 2-metre link to the beach as shown. Calculate the total distance covered in this course (starting and finishing on the beach) **PS**



10 A kite is attached to a 24 m piece of string. The rope is held 1.2 m above the ground and covers a horizontal distance of 10 m.
Find:

a the value of x, correct to one decimal place

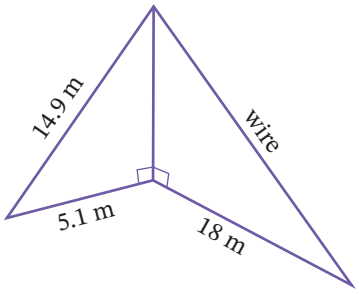
b the height of the kite above the ground, correct to the nearest metre.



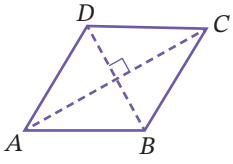
11 The diagonal of a rectangle is 200 mm in length. One of the side lengths of the rectangle is 167 mm. Find the length of the other side, correct to the nearest millimetre. **PS**

12 To keep a power pole vertically upright, it is supported by 2 wires as shown.

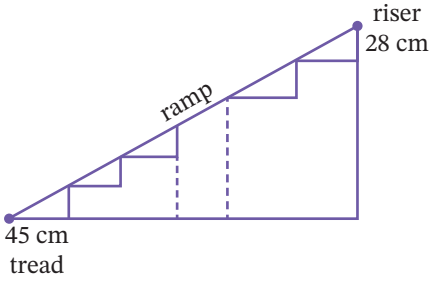
Find, correct to one decimal place, the length of the second wire. **R**



13 $DCBA$ is a rhombus with diagonals $AC = 18$ cm and $DB = 12$ cm. Find the perimeter of the rhombus, correct to one decimal place. **PS** **R**



14 A set of 24 steps is to be replaced by a ramp as shown. Each step has a tread of 45 cm and a riser of 28 cm. Find the length of the ramp in metres. **PS** **R**



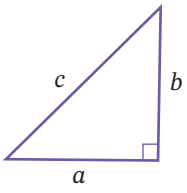
Foundation | Standard | Complex

2.08 Testing for right-angled triangles

Worksheet
Testing for right-angled triangles

Pythagoras' theorem says that if a right-angled triangle has sides of length a, b, and c, then $c^2 = a^2 + b^2$.

The reverse of this is also true:



The converse of Pythagoras' theorem

If any triangle has sides of length a, b, and c that follow the formula $c^2 = a^2 + b^2$, then the triangle must be right-angled.

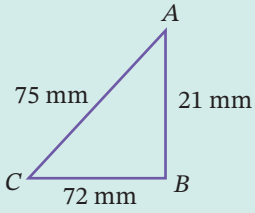
The right angle is the angle that is opposite the hypotenuse.

This is called the **converse** (or opposite) of Pythagoras' theorem, because it is the 'back-to-front' version of the theorem.

Video
Testing for right-angled triangles

Example 18

Test whether $\triangle ABC$ is right-angled.



Solution

$75^2 = 5625$	Square the longest side
$21^2 + 72^2 = 5625$	Square the 2 shorter sides, then add
$\therefore 75^2 = 21^2 + 72^2$	The sides of this triangle follow $c^2 = a^2 + b^2$
$\therefore \triangle ABC$ is right-angled.	The right angle is $\angle B$

Example 19

Rahul constructed a triangle with sides of length 37 cm, 12 cm and 40 cm. Show that these measurements do not form a right-angled triangle.

Solution

$40^2 = 1600$	Square the longest side
$12^2 + 37^2 = 1513 \neq 1600$	Square the 2 shorter sides, then add
$\therefore 40^2 \neq 12^2 + 37^2$	The sides of this triangle do not follow $c^2 = a^2 + b^2$
$\therefore$ The triangle is not right-angled.	

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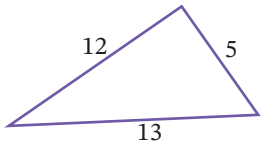
EXERCISE 2.08 ANSWERS ON P. 614

Testing for right-angled triangles U F R C

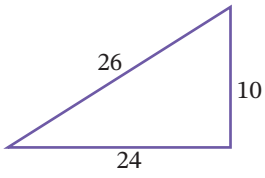
EXAMPLE 18

1 Test whether each triangle is right-angled. C

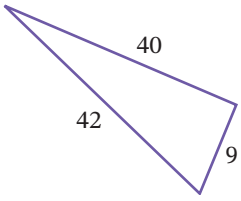
a



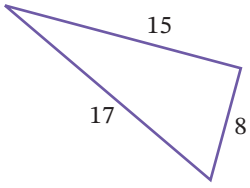
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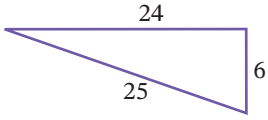
c



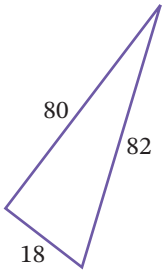
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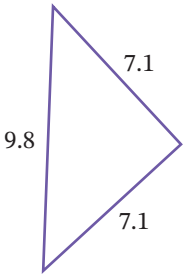
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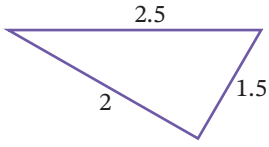
f



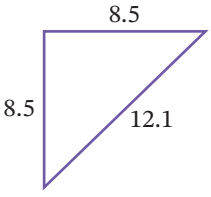
g



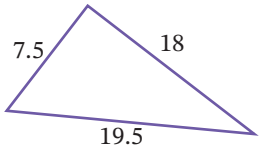
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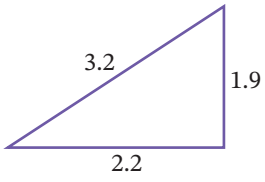
i



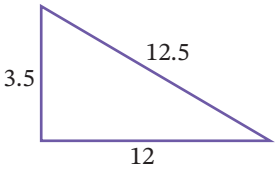
j



k



l



EXAMPLE 19

2 Paba constructed a triangle with sides of length 17 cm, 21 cm and 30 cm. Show that these measurements do *not* form a right-angled triangle. C

3 Which set of measurements would make a right-angled triangle? Select the correct answer **A**, **B**, **C** or **D**. C

A 2 cm, 3 cm, 4 cm

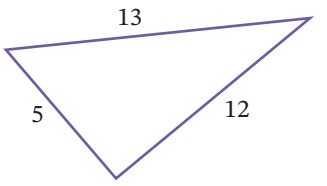
B 5 mm, 10 mm, 15 mm

C 12 cm, 16 cm, 20 cm

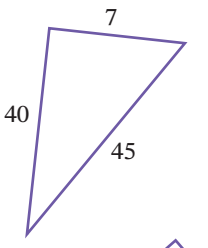
D 7 m, 24 m, 31 m

4 Which one of these triangles is not right-angled? Select **A**, **B**, **C** or **D**. C

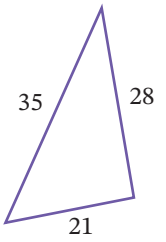
A



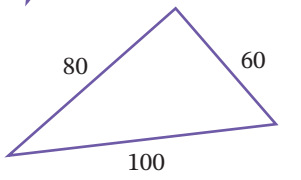
B



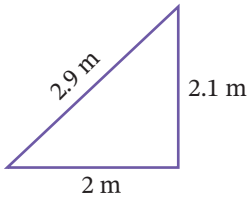
C



D



5 A pole 2.1 m tall is supported by a wire 2.9 m long. The other end of the wire is attached to the ground 2 m from the base of the pole. Is the pole standing vertically upright? R C



Quiz
Mental skills 2B

Mental skills 2B: Maths without calculators ANSWERS ON P. 614

Multiplying and dividing by a multiple of 10

1 Consider each example.

a $4 \times 700 = 4 \times 7 \times 100 = 28 \times 100 = 2800$

b $5 \times 60 = 5 \times 6 \times 10 = 30 \times 10 = 300$

c $12 \times 40 = 12 \times 4 \times 10 = 48 \times 10 = 480$

d $3.2 \times 30 = 3.2 \times 3 \times 10 = 9.6 \times 10 = 96$ (by estimation, $3 \times 30 = 90 \approx 96$)

e $4.6 \times 50 = 4.6 \times 5 \times 10 = 23 \times 10 = 230$ (by estimation, $5 \times 50 = 250 \approx 230$)

f $9.4 \times 200 = 9.4 \times 2 \times 100 = 18.8 \times 100 = 1880$ (by estimation, $9 \times 200 = 1800 \approx 1880$)

2 Now evaluate each product.

a	8×2000	**b**	3×70	**c**	11×900	**d**	2×300
e	4×4000	**f**	5×80	**g**	7×70	**h**	1.3×40
i	2.5×600	**j**	5.8×200	**k**	3.6×50	**l**	4.4×3000

Foundation Standard Complex

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3 Consider each example.

a $8000 \div 400 = 8000 \div 100 \div 4 = 80 \div 4 = 20$

b $200 \div 50 = 200 \div 10 \div 5 = 20 \div 5 = 4$

c $6000 \div 20 = 6000 \div 10 \div 2 = 600 \div 2 = 300$

d $282 \div 30 = 282 \div 10 \div 3 = 28.2 \div 3 = 9.4$

e $3520 \div 40 = 3520 \div 10 \div 4 = 352 \div 4 = 88$

f $8940 \div 200 = 8940 \div 100 \div 2 = 89.4 \div 2 = 44.7$

4 Now evaluate each quotient.

a	$560 \div 70$	**b**	$2500 \div 500$	**c**	$3200 \div 400$	**d**	$440 \div 20$
e	$160 \div 40$	**f**	$1500 \div 30$	**g**	$450 \div 50$	**h**	$744 \div 80$
i	$2550 \div 300$	**j**	$846 \div 200$	**k**	$576 \div 60$	**l**	$2160 \div 90$

Did you know?

Pythagoras and President Garfield

According to *The Guinness Book of Records*, Pythagoras' theorem has been proved the greatest number of times, with a book published in 1940 containing 370 different proofs, including one by US president James Garfield (1831–1881). Before becoming involved in politics, Garfield studied mathematics and taught in a public school. He published his proof in 1876 and was the 20th president of the United States, governing for only 150 days in 1881.

Find the main mathematical feature of President Garfield's proof.



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Pythagorean triads

2.09

A **Pythagorean triad** or **Pythagorean triple** is any set of 3 positive integers that follow Pythagoras' theorem, for example, {3, 4, 5} or {5, 12, 13}. The word **triad** means a group of 3 related items ('tri-' means 3).

Pythagorean triad

$\{a, b, c\}$ is a Pythagorean triad if $c^2 = a^2 + b^2$.

Any multiple of $\{a, b, c\}$ is also a Pythagorean triad.

Worksheet
Pythagorean triads

Technology
Pythagorean triples

Spreadsheet
Pythagorean triples

Example 20

Test whether {5, 12, 13} is a Pythagorean triad.

Solution

$13^2 = 169$ — Squaring the largest number

$5^2 + 12^2 = 169$ — Squaring the 2 smaller numbers and adding them

$\therefore 13^2 = 5^2 + 12^2$ — These 3 numbers follow Pythagoras' theorem

$\therefore$ {5, 12, 13} is a Pythagorean triad.

Example 21

{3, 4, 5} is a Pythagorean triad. Create other Pythagorean triads by multiplying (3, 4, 5) by:

a 2 **b** 9 **c** $\frac{1}{2}$

Solution

a $2 \times \{3, 4, 5\} = \{6, 8, 10\}$

Checking: $10^2 = 100$

$6^2 + 8^2 = 100$

$\therefore 10^2 = 6^2 + 8^2$

$\therefore$ {6, 8, 10} is a Pythagorean triad.

b $9 \times \{3, 4, 5\} = \{27, 36, 45\}$

Checking: $45^2 = 2025$

$27^2 + 36^2 = 2025$

$\therefore 45^2 = 27^2 + 36^2$

$\therefore$ {27, 36, 45} is a Pythagorean triad.

c $\frac{1}{2} \times \{3, 4, 5\} = \{1.5, 2, 2.5\}$

Checking: $2.5^2 = 6.25$

$1.5^2 + 2^2 = 6.25$

$\therefore 2.5^2 = 1.5^2 + 2^2$

$\therefore$ {1.5, 2, 2.5} is a Pythagorean triad.

Strictly speaking, these are not Pythagorean triads because they are not whole numbers.

EXERCISE 2.09 ANSWERS ON P. 614

Pythagorean triads U F R C

EXAMPLE 20

1 Test whether each triad is a Pythagorean triad.

a {8, 15, 17} **b** {10, 24, 26} **c** {30, 40, 50}

d {5, 7, 9} **e** {9, 40, 41} **f** {4, 5, 9}

g {11, 60, 61} **h** {7, 24, 25} **i** {15, 114, 115}

2 Which of the following is a Pythagorean triad? Select the correct answer **A**, **B**, **C** or **D**.

A {4, 6, 8} **B** {5, 10, 12} **C** {6, 7, 10} **D** {20, 48, 52}

3 Use the spreadsheet you created in **Technology: Finding the hypotenuse** on page 58 to check your answers to questions **1** and **2**.



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EXAMPLE 21

4 For each Pythagorean triad, create another triad by multiplying each number in the triad by:

i a whole number **ii** a fraction **iii** a decimal.

a {5, 12, 13} **b** {8, 15, 17} **c** {30, 40, 50} **d** {7, 24, 25}

Check that each answer follows Pythagoras' theorem.

5 Pythagoras developed formulas for finding Pythagorean triads $\{a, b, c\}$. If one number in the triad is a, the formulas for the other 2 numbers are $b = \frac{1}{2}(a^2 - 1)$ and $c = \frac{1}{2}(a^2 + 1)$.

a If $a = 5$, use the formulas to find the values of b and c.

b Show that $\{a, b, c\}$ from part **a** is a Pythagorean triad.

6 Use the formulas to find Pythagorean triads for each value of a.

a $a = 7$ **b** $a = 11$ **c** $a = 15$ **d** $a = 4$

e $a = 9$ **f** $a = 19$ **g** $a = 10$ **h** $a = 51$

7 Prove that if $b = \frac{1}{2}(a^2 - 1)$ and $c = \frac{1}{2}(a^2 + 1)$, then $c^2 = a^2 + b^2$ is true. R C

8 There are many other formulas for creating Pythagorean triads. Use the Internet to research some of them.

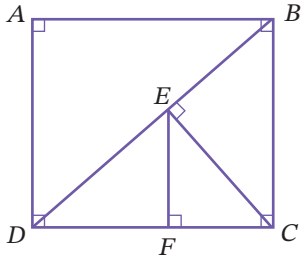
Power plus ANSWERS ON P. 614

1 Simplify each expression as an exact value.

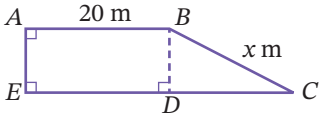
a $\left(\sqrt{2}\right)^3$ **b** $\left(\sqrt{2}\right)^4$ **c** $\left(-\sqrt{3}\right)^4$

d $\left(-\sqrt{3}\right)^3$ **e** $\left(2\sqrt{5}\right)^3$ **f** $\left(-3\sqrt{6}\right)^4$

2 Name the 5 hypotenuses on this diagram.



3 The rectangle $ABDE$ in the diagram has an area of 160 m^2. Find the value of x as a surd if $\triangle BDC$ is an isosceles triangle.

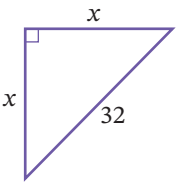


2.09

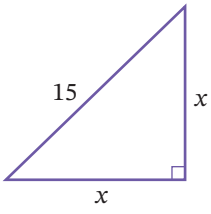
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4 Find the exact value of x in each triangle.

a



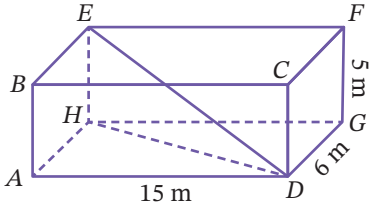
b



5 For this rectangular prism, find, correct to one decimal place, the length of diagonal:

a HD

b DE

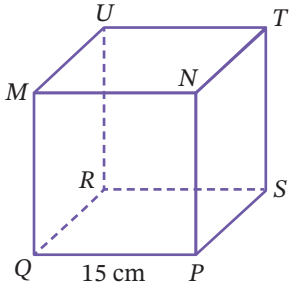


6 Find, correct to one decimal place, the length of the interval joining points $A(-2, 1)$ and $B(3, -2)$ on the number plane.

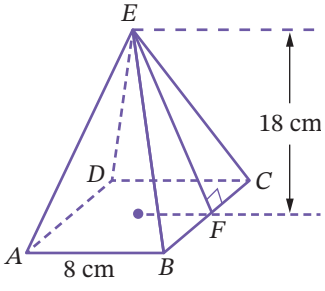
7 For this cube, find, correct to 2 decimal places, the length of diagonal:

a QS

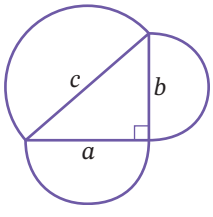
b QT



8 For this square pyramid, find, as a surd, the slant height EF.



9 Use Pythagoras' theorem to prove that for a right-angled triangle, the area of the semicircle constructed on the hypotenuse is equal to the areas of the semicircles constructed on the other 2 sides.



CHAPTER 2 REVIEW

Language of maths

area	converse	diagonal	formula
hypotenuse	irrational	perimeter	Pythagoras
right-angled	shorter	side	square root
surd	theorem	triad	unknown

1 State Pythagoras' theorem in words.

2 Describe the **hypotenuse** of a right-angled triangle in 2 ways.

3 What is another word for 'theorem'?

4 For what type of triangle is **Pythagoras' theorem** used?

5 What is a **surd**?

6 What is the name given to a set of 3 numbers that follows Pythagoras' theorem?

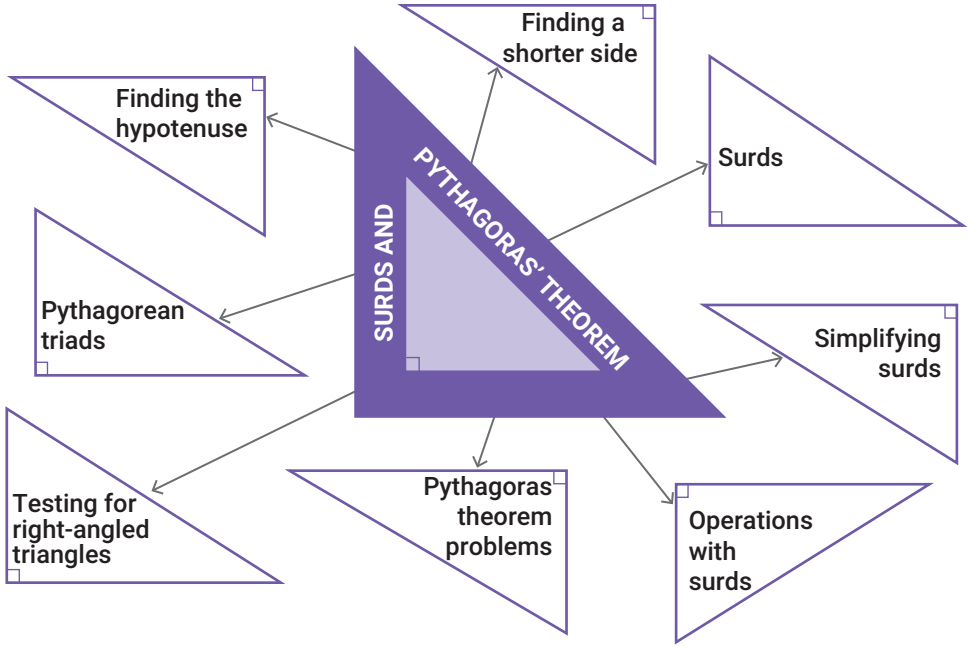
Quiz
Language of maths 2

Puzzle
Pythagoras' theorem find-a-word

Topic summary

- How relevant do you think Pythagoras' theorem is to our world? Give reasons for your answer.
- Give 3 examples of jobs where Pythagoras' theorem would be used.
- What did you find especially interesting about this topic?
- Is there any section of this topic that you found difficult? Discuss any problems with your teacher or a friend.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Surds and Pythagoras' theorem

Quiz
Test yourself 2

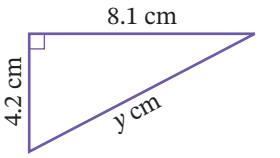
TEST YOURSELF 2

ANSWERS ON P. 615

2.01

1 Find the value of y. Select the closest answer **A**, **B**, **C** or **D**.

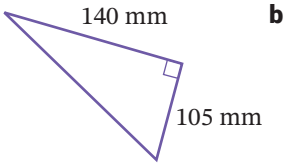
A 12.3 **B** 9.1
C 6.9 **D** 3.5



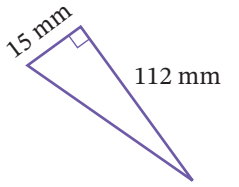
2.01

2 For each triangle, find the length of the hypotenuse. Give your answer as an exact value.

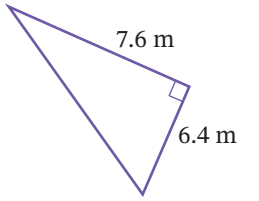
a



b



c



2.02

3 Is each number rational (R) or irrational (I)?

a $\sqrt{8}$ **b** $\frac{22}{7}$ **c** $0.5\dot{7}$ **d** $3\sqrt{5}$ **e** $\sqrt{81}$
f $\sqrt[3]{125}$ **g** $\sqrt[3]{-8}$ **h** $0.\dot{1}2\dot{3}$ **i** $5+\sqrt{3}$

PATH

2.03

4 Simplify each surd.

a $\sqrt{72}$ **b** $\sqrt{98}$ **c** $\sqrt{275}$ **d** $\sqrt{128}$ **e** $3\sqrt{150}$
f $7\sqrt{28}$ **g** $4\sqrt{288}$ **h** $5\sqrt{45}$ **i** $7\sqrt{48}$

2.04

5 Simplify each expression.

a $8\sqrt{5}-\sqrt{5}$ **b** $\sqrt{7}-3\sqrt{2}+2\sqrt{7}$
c $6+\sqrt{5}-4+5\sqrt{5}$ **d** $7\sqrt{7}-\sqrt{7}-3\sqrt{7}$
e $\sqrt{200}+\sqrt{18}$ **f** $3\sqrt{5}+\sqrt{50}-2\sqrt{125}$
g $7\sqrt{32}-\sqrt{27}-2\sqrt{98}+4\sqrt{75}$ **h** $4\sqrt{45}-3\sqrt{63}+5\sqrt{80}$

2.05

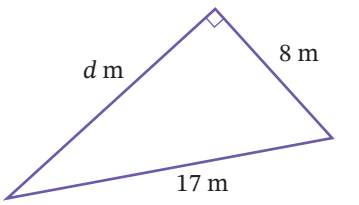
6 Simplify each expression.

a $\sqrt{3}\times\sqrt{7}$ **b** $\sqrt{8}\times\sqrt{5}$ **c** $\sqrt{6}\times\sqrt{8}$
d $\sqrt{5}\times\sqrt{11}$ **e** $\sqrt{8}\times\sqrt{3}$ **f** $\sqrt{98}\div\sqrt{7}$
g $8\sqrt{42}\div2\sqrt{7}$ **h** $\sqrt{125}\div5\sqrt{5}$ **i** $4\sqrt{147}\div12\sqrt{98}$
j $\frac{\sqrt{18}\times\sqrt{3}}{\sqrt{12}}$ **k** $\frac{\sqrt{6}\times\sqrt{24}}{\sqrt{27}\times2\sqrt{3}}$

2.06

7 Find the value of d. Select **A**, **B**, **C** or **D**.

A 12 m **B** 15 m
C 16.8 m **D** 18.9 m

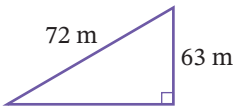


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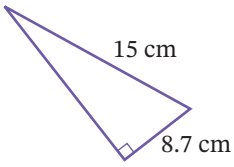
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8 For each triangle, find the length of the unknown side, correct to one decimal place. 2.06

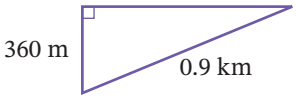
a



b

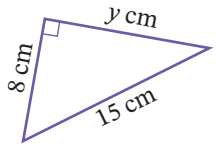


c

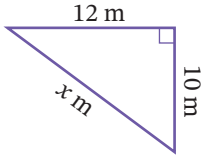


9 Find the exact value of the variable in each triangle. 2.07

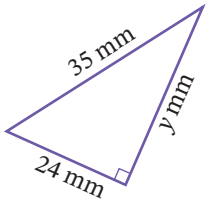
a



b



c



10 In $\triangle DEF$, $\angle F = 90°$, $DF = 84$ cm, and $EF = 1.45$ m. Find DE, correct to the nearest centimetre. 2.07

11 Julie walks 6 km due east from a starting point P, while Luka walks 4 km due south from P. 2.07

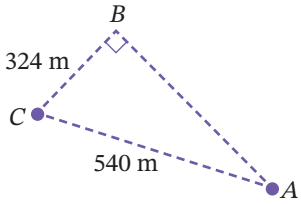
a Draw a diagram showing this information.

b How far are Julie and Luka apart? Give your answer correct to one decimal place.

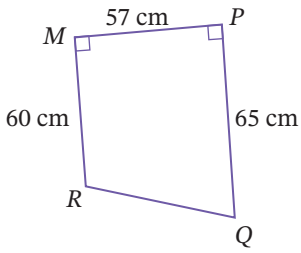
12 On a cross-country ski course, checkpoints C and A are 540 m apart and checkpoints C and B are 324 m apart. Find the distance: 2.07

a between checkpoints A and B

b skied in one lap of the course in kilometres.

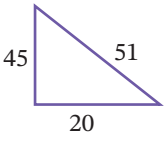


13 Find the perimeter of this trapezium, correct to one decimal place. 2.07

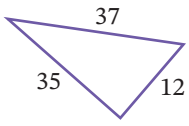


14 Test whether each triangle is right-angled. 2.08

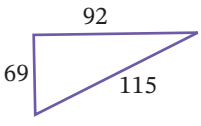
a



b



c



15 Test whether each triad is Pythagorean. 2.09

a {7, 24, 25} **b** {5, 7, 10} **c** {20, 21, 29} **d** {1.1, 6, 6.1}

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